

	Age of wine x	Price of wine y	Predicted price of wine $\hat{y} = B_0 + B_1x$	Residuals ($e_y = y - \hat{y}$)	Squared residuals $(y - \hat{y})^2$
1	0	75	$B_0 + B_1 * 0$	$75 - (B_0 + B_1 * 0)$	$[75 - (B_0 + B_1 * 0)]^2$
2	1.2	100	$B_0 + B_1 * 1.2$	$100 - (B_0 + B_1 * 1.2)$	$[100 - (B_0 + B_1 * 1.2)]^2$
3	2.1	220	$B_0 + B_1 * 2.1$	$220 - (B_0 + B_1 * 2.1)$	$[220 - (B_0 + B_1 * 2.1)]^2$
4	3.4	300	$B_0 + B_1 * 3.4$	$300 - (B_0 + B_1 * 3.4)$	$[300 - (B_0 + B_1 * 3.4)]^2$
5	4.1	440	$B_0 + B_1 * 4.1$	$440 - (B_0 + B_1 * 4.1)$	$[440 - (B_0 + B_1 * 4.1)]^2$
6	5.6	500	$B_0 + B_1 * 5.6$	$500 - (B_0 + B_1 * 5.6)$	$[500 - (B_0 + B_1 * 5.6)]^2$
7	7	700	$B_0 + B_1 * 7$	$700 - (B_0 + B_1 * 7)$	$[700 - (B_0 + B_1 * 7)]^2$
Sum	23.4	2335	$7B_0 + 16.4B_1$	$2335 - (7B_0 + 23.4B_1)$	Se
Average	3.34 is $\bar{x}$	333.57 is $\bar{y}$	$B_0 + 3.34B_1$ is $B_0 + B_1 \bar{x}$	$333.57 - (B_0 + 3.34B_1)$ is $\bar{y} - (B_0 + \bar{x}B_1)$	Se / 7

Table 3

The R value is 0.985, which is close to 1. Hence, the two variables are strongly positively correlated

Note: In simple terms, the shape of the data is important to know for selecting the model for prediction. So always draw the plot to get a sense of the data.

Linear regression equation

Now that we have seen that our data is a good use case for linear regression, let's have a look at the formula. The linear equation is:

$$y = B_0 + B_1 * x$$

Here, y is the predicted variable. B_0 is the intercept — the predicted value of y when x is 0. In this example, you can see that when x is 0, the value of y is 75. B_1 is the regression coefficient, which refers to the change expected in y as x increases. Again going back to the example, as the number of years increases, the price of the wine also increases. Now, let's try and find out how much it increases.

$$y = B_0 + B_1 * x_1 + B_2 * x_2 + \dots + B_N * x_N$$

When we have multiple input variables, as seen here, the aim of the linear regression algorithm is to find the best values for $B_0, B_1, \dots, B_N$.

A model 'train, validate and test' set

Before training our model, we need to do a train and validation split.

- **Train:** Set of input and output data for the model to learn.
- **Validate:** Set of input and output data used for evaluation of the model, to see it is not over fitted and reacting well to unseen data. It is also used for fine tuning hyper parameters, feature selection, threshold cut-off selection, and so on.
- **Test:** Final set of unseen data to evaluate the final model.

OLS calculation

In this method, we have to choose the values of B_0 and B_1 such that the total sum of squares of the difference between the calculated and observed values of y is minimised.

Before we write the source code in Python, we need to understand how the OLS works if we had to do things manually.

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

Figure 4: Correlation coefficient formula
(Credits: <https://www.statisticshowto.com/probability-and-statistics/correlation-coefficient-formula/#Pearson>)

Step 1: Calculate the data given in Table 2 from the formulas given below, i.e., S_{xx} – Sum of squares of x; S_{yy} – Sum of squares of y; and S_{xy} – Sum of products of x and y.

The formulas are:

- $\bar{x}$ – Mean of x
- $\bar{y}$ – Mean of y
- S_{xx} – Sum of squares of $(x - \bar{x})$
- S_{yy} – Sum of squares of $(y - \bar{y})$
- S_{xy} – Sum of products of $(x - \bar{x})$ and $(y - \bar{y})$

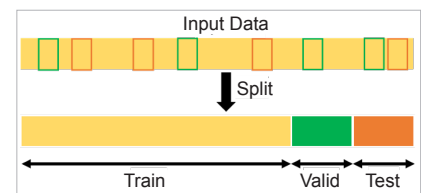


Figure 5: Randomly split the input data into a train, valid, and test set (Credits: <https://towardsdatascience.com/how-to-split-data-into-three-sets-train-validation-and-test-and-why-e50d22d3e54c>)